SoccerGMR Performance Benchmark: Baseline vs. SI-Bind (Ours - E6)

Head-to-head evaluation across 9 temporal DETR backbones on Standard Test (1,036 queries) and Full-Decodable Test (2,970 queries). Strict single seed (2023).

		Standard Test Split (1,036 queries)				Full-Decodable Test Split (2,970 queries)			
Backbone	**Configuration**	**mAP (%)**	**Orc-Gap (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**	**mAP (%)**	**Orc-Gap (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**
Moment-DETR	1. GMR Baseline	6.68	24.28	**67.94**	70.20	5.47	23.48	**72.08**	78.91
	2. GMR + SI-Bind (Ours - E6)	**15.21**	**23.66**	65.52	**74.32**	**12.58**	**22.21**	68.83	**81.86**
EaTR	1. GMR Baseline	16.49	**21.51**	65.60	**76.56**	15.19	**20.39**	67.38	**81.59**
	2. GMR + SI-Bind (Ours - E6)	**17.12**	21.80	**67.74**	76.23	**15.35**	20.89	**68.66**	81.03
QD-DETR	1. GMR Baseline	14.62	**20.23**	71.48	**77.01**	12.65	**19.05**	72.49	81.75
	2. GMR + SI-Bind (Ours - E6)	**16.30**	20.65	**73.71**	76.63	**13.20**	19.84	**75.91**	**82.47**
CG-DETR	1. GMR Baseline	**18.65**	25.66	57.47	**76.40**	14.67	23.28	63.23	**81.81**
	2. GMR + SI-Bind (Ours - E6)	17.38	**22.89**	**70.85**	75.92	**15.54**	**21.61**	**73.14**	80.94
BM-DETR	1. GMR Baseline	4.00	**17.59**	**75.00**	**71.55**	3.65	**17.62**	**78.58**	79.11
	2. GMR + SI-Bind (Ours - E6)	**7.72**	20.91	73.40	70.18	**7.03**	20.30	77.47	**79.27**
BAM-DETR	1. GMR Baseline	23.25	**19.82**	**70.43**	**78.66**	20.62	**18.37**	**71.79**	**80.99**
	2. GMR + SI-Bind (Ours - E6)	**23.78**	20.40	68.96	78.37	**21.78**	19.07	69.28	78.69
Sim-DETR	1. GMR Baseline	20.73	22.51	74.69	72.97	19.18	21.78	**75.72**	**78.87**
	2. GMR + SI-Bind (Ours - E6)	**21.17**	**21.26**	**74.89**	**75.12**	**20.08**	**20.83**	73.10	77.83
UVCOM	1. GMR Baseline	7.12	17.24	75.00	70.03	6.40	16.83	**76.80**	**78.98**
	2. GMR + SI-Bind (Ours - E6)	**18.99**	**16.30**	**75.40**	**76.91**	**17.74**	**15.55**	73.11	75.20
FlashVTG	1. GMR Baseline	**25.43**	**25.47**	**66.35**	**74.30**	**20.43**	**24.13**	**68.48**	**82.21**
	2. GMR + SI-Bind (Ours - E6)		*-- running / pending --*				*-- running / pending --*		

Method 1 (GMR Baseline): Official temporal DETR baseline checkpoints evaluated with native bipartite matching.
Method 2 (GMR + SI-Bind [Ours - E6]): Full Structural Triad (Temporal Residual Stream + Slot-Private Innovation + Dominance-Aware Refresh).
Orc-Gap (%): Area between Oracle envelope and G-mIoU@1 threshold curve across tau in [0, 1] (Integral of Oracle_G1 minus G1-AUC). Lower is better.
Bold values highlight the optimal performance between Baseline and SI-Bind (E6) per backbone. FlashVTG E6 currently in queue.